One-dimensional Turbulence Model (notes)

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Governing equations for flow

Reynolds-averaged Navier Stokes

The Reynolds-Averaged Navier Stokes reads:

$$\frac{\partial \overline{u_i}}{\partial t} + \overline{u_j}\frac{\partial \overline{u_i}}{\partial x_j} = \frac{\partial}{\partial x_k}\overline{u_k u_i} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x_i} - g\delta_{i3} + \nu\frac{\partial^2}{\partial x_j \partial x_j}\overline{u_i} \tag{1}$$

where flow is considered to be divergent-free, or solenoidal,

$$\frac{\partial u_i}{\partial x_i} = 0. \tag{2}$$

The Boussinesq approximation is invoked to represent that density fluctuaions affect bouyancy but no other transport/state terms. These fluctuations are written $\beta\theta$, where β is the coefficient of thermal expansion. This equation in the $\hat{e_1}$ direction, expanded, reads

$$\frac{\partial \overline{u}}{\partial t} + \overline{u}\frac{\partial \overline{u}}{\partial x} + \overline{v}\frac{\partial \overline{u}}{\partial y} + \overline{w}\frac{\partial \overline{u}}{\partial z} = \frac{\partial}{\partial x}\overline{u'u'} + \frac{\partial}{\partial y}\overline{v'u'} + \frac{\partial}{\partial z}\overline{w'u'} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right).$$

We neglect advection and gradients in the x and y directions and take $w = 0$ as established earlier,

$$\frac{\partial \overline{u}}{\partial t} + \overline{u}\cancel{\frac{\partial \overline{u}}{\partial x}} + \overline{v}\cancel{\frac{\partial \overline{u}}{\partial y}} + \overline{w}\cancel{\frac{\partial \overline{u}}{\partial z}} = \cancel{\frac{\partial}{\partial x}\overline{u'u'}} + \cancel{\frac{\partial}{\partial y}\overline{v'u'}} + \frac{\partial}{\partial z}\overline{w'u'} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu\left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \cancel{\frac{\partial^2 u}{\partial y^2}} + \frac{\partial^2 u}{\partial z^2}\right)$$

We now have a simplified equation for Reynolds-averaged flow in the $\hat{e_1}$ direction.

$$\frac{\partial \overline{u}}{\partial t} = \frac{\partial}{\partial z}\overline{w'u'} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu\frac{\partial^2 u}{\partial z^2}.$$

Next, we apply the Boussinesq assumption, which allows us to re-write the Reynolds Stress as a function of the mean velocity gradient. Note that the k term drops out as our Reynolds stress is off-diagonal,

$$\overline{w'u'} = \mu_t\left(\frac{\partial \overline{u}}{\partial z} + \frac{\partial \overline{w}}{\partial x}\right) - \cancel{\frac{2}{3}k\delta_{13}} = \mu_t\left(\frac{\partial \overline{u}}{\partial z} + \cancel{\frac{\partial \overline{w}}{\partial x}}\right) = \mu_t\frac{\partial \overline{u}}{\partial z}$$

We plug in our expression for the Reynold's stress,

$$\frac{\partial \overline{u}}{\partial t} = \frac{\partial}{\partial z}\left(\mu_t\frac{\partial \overline{u}}{\partial z}\right) - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu\frac{\partial^2 \overline{u}}{\partial z^2}$$

$$\frac{\partial \overline{u}}{\partial t} = \mu_t \frac{\partial^2 \overline{u}}{\partial z^2} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu \frac{\partial^2 \overline{u}}{\partial z^2}$$

We can group the turbulent and molecular viscosity terms,

$$\frac{\partial \overline{u}}{\partial t} = -\frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x}(\nu_t + \nu)\frac{\partial^2 \overline{u}}{\partial z^2}$$

however, since we assume that turbulent viscosity is far greater than molecular viscosity,

$$\nu_t >> \nu$$

we often neglect molecular viscosity, leaving us with

$$\frac{\partial \overline{u}}{\partial t} = -\frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu_t \frac{\partial^2 \overline{u}}{\partial z^2}.$$

After applying the Boussinesq assumption, we've escaped the need to resolve the Reynold's Stress, however, in discretizing this equation, we still need to approximate this turbulent viscosity ν_t. Hence the need for a closure scheme.

Turbulence closure

Similarity theory tells us that ν_t, which has units of [length^2 time^{-1}] can be scaled by a length scale times a velocity scale,

$$\nu_t \approx \ell_T \; U_T$$

Here I am using the subscript T to denote these quanitities being *turbulent scales.* We estimate the velocity scale using the square root of the total turbulent kinetic energy,

$$\mathrm{TKE} = \mathrm{tr}(\overline{u_i u_j}) = (\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$$

$$q = \sqrt{\overline{u'^2} + \overline{v'^2} + \overline{w'^2}}$$

$$\nu_t \approx \ell_T \; q$$

Again, ν_t is a function $\nu_t(z,t)$ that dissipates energy from the mean velocity (due to subscale turbulence).

The TKE equation

We can derive an equation for the turbulent kinetic energy, q^2. This process is fairly intensive so I will put it in an appendix or something. The equation for the turbulent kinetic energy q^2 given by,

$$\underbrace{\frac{\partial q^2}{\partial t}}_{1} = \underbrace{\frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z})}_{2} + \underbrace{2P}_{2} + \underbrace{2B}_{3} - \underbrace{2\epsilon}_{4}$$

Moving from left to right,

$$[1] \quad \frac{\partial q^2}{\partial t}$$

is **unsteadiness**, or the change in turbulent kinetic energy over time;

$$[2] \quad \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z})$$

is the **turbulent diffusion**, or a measure of how turbulent energy "diffuses" throughout the fluid flow. This neither creates nor destroys turbulence– just moves it. Next is

$$[3] \quad P = \overline{u_i u_k} \frac{\partial u_i}{\partial x_k},$$

production, also called **shear production**. We expand this for our simplified scenario (neglecting gradients of x, y, $w = 0$)

$$P = \overline{u'w'} \frac{\partial u}{\partial z} + \overline{v'w'} \frac{\partial v}{\partial z} + \overline{w'w'} \cancel{\frac{\partial w}{\partial z}}$$

$$P = \mu_t \left(\frac{\partial \overline{u}}{\partial z} + \cancel{\frac{\partial \overline{w}}{\partial z}} \right) \frac{\partial u}{\partial z} + \mu_t \left(\frac{\partial \overline{v}}{\partial z} + \cancel{\frac{\partial \overline{w}}{\partial y}} \right) \frac{\partial v}{\partial z}$$

$$P = \mu_t \left(\frac{\partial \overline{u}}{\partial z} \right)^2 + \mu_t \left(\frac{\partial \overline{v}}{\partial z} \right)^2$$

Dissipation is a disaster

$$[4] \quad \epsilon = -\nu \left(\overline{\frac{\partial u_i}{\partial x_k} \frac{\partial u_i}{\partial x_k}} \right)$$

$$\epsilon = -\nu \left(\overline{\left(\frac{\partial u}{\partial z} \right)^2} + \overline{\left(\frac{\partial v}{\partial z} \right)^2} + \overline{\left(\cancel{\frac{\partial w}{\partial z}} \right)^2} \right)$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) + \nu_t \frac{\partial u}{dz}^2 + 2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z} - 2 \frac{(q^2)^{3/2}}{B_1 l}$$

Similar to our simplications before, we can comfortably neglect advection of TKE as long as the time scale of dissipation is shorter than the advection time scale T_a over one grid cell in the horizontal.

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) - \overline{u_i u_k}\frac{\partial u_i}{\partial x_k} + 2B - \nu \overline{\frac{\partial u_i}{\partial x_k}\frac{\partial u_i}{\partial x_k}}$$

Boundary layer approximation for environmental flow

The equations used in our model assume that in our environment, the lateral length scales L_x and L_y are far greater than the vertical length scale (or depth) L_z. This means that gradients in the x and y directions can be seen as very small, or negligble. We maintain gradients in the pressure direction as these drive the flow and are assumed not to be functions of x.

$$\frac{\partial u}{\partial t} + u\cancel{\frac{\partial u}{\partial x}} + v\cancel{\frac{\partial u}{\partial y}} + w\frac{\partial z}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu_t\left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \cancel{\frac{\partial^2 u}{\partial y^2}} + \frac{\partial^2 u}{\partial z^2}\right)$$

$$\frac{\partial u}{\partial t} + w\frac{\partial z}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu_t\left(\frac{\partial^2 u}{\partial z^2}\right)$$

We also can prove that $w = 0$, simplifying our equation to

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu_t\left(\frac{\partial^2 u}{\partial z^2}\right) \tag{3}$$

Boussinesq gradient diffusion hypothesis

Boussinesq attempted to simplify the mess of the Reynold's stress tensor $\overline{u_i u_j}$ by reasoning that these correlation terms would act like a stress tensor in a flow.

In deriving the Navier-Stokes we define a stress tensor as

$$-(\tau_{ij} + P\delta_{ij}) = -2\mu S_{ij} = -\mu\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i})\right)$$

where τ_{ij} represents shear stresses and $P\delta_{ij}$, or pressure, acts as a body force along the diagonal of the tensor.

If the Reynold's stresses are like shear stresses imposed on flow, we can subsitute them in here

$$-(\overline{u_i u_j} + \frac{2}{3}k\delta_{ij}) = -\mu_t\left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i}\right)$$

only this turbulent stress tensor would be approximated by some turbulent viscosity μ_t and the

$$\overline{u_i u_j} = \mu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}$$

Note that the trace of the RHS would be

$$\mu_t (2\frac{\partial u_i}{\partial x_i} - \frac{2}{3}k) = \mu_T(-3\frac{2}{3}k) = -2\mu_T k = -2\mu_T(\frac{1}{2}\overline{u_i' u_i'})$$

The main thing to note here is that using this approxmiation, we can now represent the Reynold's stress as being driven by gradients in the mean flow. This will help us from a closure perspective.

The two-equation model

Our equation for the turbulent kinetic energy q is now given by,

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) + 2P + 2B - 2\epsilon$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) - \overline{u_i u_k}\frac{\partial u_i}{\partial x_k} + 2B - \nu \frac{\partial u_i}{\partial x_k}\frac{\partial u_i}{\partial x_k}$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) + \nu_t \frac{\partial u}{dz}^2 + 2\frac{g}{\rho_0}\kappa_p \frac{\partial p}{\partial z} - 2\frac{(q^2)^{3/2}}{B_1 l}$$

We can neglect advection as long as the time scale of dissipation ir short than the advection time scale T_a over one grid cell in the horizontal.

The Mellor-Yamada 2.5 closure

The chosen closure scheme for our 1-D turbulent model is the Mellor-Yamada 2.5 scheme. This scheme approximates the eddy viscosity as

$$\nu_t = S_m q L \tag{4}$$

and the turbulent diffusivity as

$$\kappa_q = S_h Q L, \tag{5}$$

where S_m and S_h are empircally-derived stability functions, q is the turbulent kinetic energy, and l is our turbulent length scale.

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}\left(qlS_q\frac{\partial q^2}{\partial z}\right) + 2\left(P_s + P_b - \frac{q^3}{B_1 l}\right) \tag{6}$$

$$\frac{\partial q^2 l}{\partial t} = \frac{\partial}{\partial z}\left(qlS_q\frac{\partial q^2 l}{\partial z}\right) + lE_1(P_s + P_b) - \frac{q^3}{B_1 l}W(z) \tag{7}$$

where P_s denotes shear production, and is given by

$$P_s = \nu_T\left(\left(\frac{\partial U}{\partial z}\right)^2 + \left(\frac{\partial V}{\partial z}\right)^2\right),$$

the buoyancy production, P_b, is calculated as

$$P_b = -\kappa_t \frac{g}{\rho_0}\frac{\partial \overline{\rho}}{\partial z},$$

and finally the turbulent dissipation is given by

$$\epsilon = \frac{q^3}{B_1 l}.$$

The function $W(z)$ is a wall-proximity function that adjusts dissipation as z approaches a boundary. S_q and E_1 are both empirical constants.

Discretization

Advection-Diffusion Equation for Algae

Our governing equation for algae is given by,

$$\frac{\partial C}{\partial t} + w_s\frac{\partial C}{\partial z} = \frac{\partial}{\partial z}\left(\kappa_t\frac{\partial C}{\partial z}\right) \tag{8}$$

where w_s is the sinking or swimming speed of the algae – it can be positive (*Microcystis*) or negative (diatoms).

We break down the discretization of Equation 8 by the left hand (time derivative + advection) and right hand (diffusion) sides. The right-hand side is trickiest.

Discretizing diffusion

We evaluate the diffusion term first in space. Note here that diffusion is given by κ_t, which respresents turbulent diffusion (a quantity we must solve for). Because the subscripts get messy, I'll drop the subscript t, but all diffusion in this equation refers to κ_t.

$$\frac{\partial}{\partial z}\left(\kappa_t \frac{\partial C}{\partial z}\right) \rightarrow \frac{1}{\Delta z}\left(\left(\kappa_t \frac{\partial C}{\partial z}\right)_{i+1/2} - \left(\kappa_t \frac{\partial C}{\partial z}\right)_{i-1/2}\right)$$

We represent these 1/2 stencil points as the average of the two neighboring cells (this is where I will now drop the subscript on κ_t),

$$\kappa_{i+1/2} \approx \frac{\kappa_{i+1} + \kappa_i}{2}$$

$$\kappa_{i-1/2} \approx \frac{\kappa_{i-1} + \kappa_i}{2}$$

And same for concentration,

$$\left(\frac{\partial C}{\partial z}\right)_{i+1/2} \approx \frac{C_{i+1} - C_i}{\Delta z}$$

$$\left(\frac{\partial C}{\partial z}\right)_{i-1/2} \approx \frac{C_i - C_{i-1}}{\Delta z}$$

Putting these together,

$$\frac{1}{\Delta z}\left(\left(\kappa \frac{\partial C}{\partial z}\right)_{i+1/2} - \left(\kappa \frac{\partial C}{\partial z}\right)_{i-1/2}\right) \rightarrow \frac{1}{\Delta z}\left(\frac{\kappa_{i+1} + \kappa_i}{2}\frac{C_{i+1} - C_i}{\Delta z} - \frac{\kappa_{t(i-1)} + \kappa_i}{2}\frac{C_i - C_{i-1}}{\Delta z}\right)$$

$$= \frac{1}{2\Delta z^2}\left((\kappa_{i+1} + \kappa_i)(C_{i+1} - C_i) - (\kappa_{i-1} + \kappa_i)(C_i - C_{i-1})\right)$$

Factor according to C,

$$\begin{aligned} &= C_{i+1}\left(\frac{1}{2\Delta z^2}\left(\kappa_{i+1} + \kappa_i\right)\right) \\ &+C_i\left(\frac{1}{2\Delta z^2}\left(-\kappa_{i+1} - 2\kappa_i - \kappa_{i-1}\right)\right) \\ &+C_{i-1}\left(\frac{1}{2\Delta z^2}\left(-\kappa_{i-1} - \kappa_i\right)\right) \end{aligned}$$

Discretizing in time

Next we discretize

$$\frac{\partial C}{\partial t} + w_s \frac{\partial C}{\partial z} = \frac{\partial}{\partial z}\left(\kappa_t \frac{\partial C}{\partial z}\right)$$

in time. We use an implicit scheme,

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} + w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = \left(\kappa_t \frac{\partial C}{\partial z}\right)^{n+1}$$

We plug in our expression for the Laplacian discretized in space, evaluated at time $n+1$,

$$\begin{aligned}
\frac{C_i^{n+1} - C_i^n}{\Delta t} + w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = C_{i+1}^{n+1}\left(\frac{1}{2\Delta z^2}\left(\kappa_{i+1}^n + \kappa_i\right)\right) \\
+C_i^{n+1}\left(\frac{1}{2\Delta z^2}\left(-\kappa_{i+1}^n - 2\kappa_i^n - \kappa_{i-1}^n\right)\right) \\
+C_{i-1}^{n+1}\left(\frac{1}{2\Delta z^2}\left(-\kappa_{i-1}^n - \kappa_i^n\right)\right)
\end{aligned}$$

$$\begin{aligned}
C_i^{n+1} - C_i^n + \Delta t w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = C_{i+1}^{n+1}\left(\frac{\Delta t}{2\Delta z^2}\left(\kappa_{i+1}^n + \kappa_i^n\right)\right) \\
+C_i^{n+1}\left(\frac{\Delta t}{2\Delta z^2}\left(-\kappa_{i+1}^n - 2\kappa_i^n - \kappa_{i-1}^n\right)\right) \\
+C_{i-1}^{n+1}\left(\frac{\Delta t}{2\Delta z^2}\left(-\kappa_{i-1}^n - \kappa_i^n\right)\right)
\end{aligned}$$

This all simplifies out to,

$$\begin{aligned}
C_i^n = C_{i+1}^{n+1}\left(-\frac{\Delta t}{2\Delta z^2}(\kappa_i^n + \kappa_{i+1}^{n+1})\right) \\
+C_i^{n+1}\left(1 + \gamma_i^n + \frac{\Delta t w_s}{\Delta z} + \frac{\Delta t}{2\Delta z^2}(\kappa_{i+1}^n + 2\kappa_i^n + \kappa_{i-1}^n)\right) \\
+C_{i-1}^{n+1}\left(-\frac{\Delta t w_s}{\Delta z} - \frac{\Delta t}{2\Delta z^2}(\kappa_{i-1}^n + \kappa_i^n)\right)
\end{aligned}$$

```
ws = - 0.002 # settling velocity
dtdz = (dt/dz)
gamma = np.ones(N-1)* 0 # Growth + loss term !!!
```

```
aA[1:top]  = - beta/2 * (Kzp[0:top-1]+ Kzp[1:top])
bA[1:top]  =  1 + ws*dtdz  + beta/2 *(Kzp[2:top+1] + 2*Kzp[1:top] + Kzp[0:top-1])
cA[1:top] = -ws*dtdz  - beta/2 * (Kzp[1:top] + Kzp[2:top+1])
dA = Ap

# Bottom-Boundary: no flux for scalars
bA[0] = 1  + beta/2 * (Kzp[1] + Kzp[0]) # + ws*dtdz
cA[0] = -beta/2 * (Kzp[1] + Kzp[0])
dA[0] =  Ap[0]

# Top-Boundary: no flux for scalars
aA[top] = -ws*dtdz - beta/2 * (Kzp[top] + Kzp[top-1])
bA[top] = 1 + ws*dtdz  + beta/2 * (Kzp[top] + Kzp[top-1])
dA[top] = Ap[top]
```

Turbulent kinetic energy equation

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) + 2P + 2B - 2\epsilon$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) + \nu_t \frac{\partial u}{\partial z}^2 + 2\frac{g}{\rho_0}\kappa_p \frac{\partial p}{\partial z} - 2\frac{(q^2)^{3/2}}{B_1 l}$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) + \underbrace{\nu_t \left(\frac{\partial u}{\partial z}\right)^2}_{\text{shear production}} + \underbrace{2\frac{g}{\rho_0}\kappa_p \frac{\partial p}{\partial z}}_{\text{bouyancy production}} - \underbrace{2\frac{(q^2)^{3/2}}{B_1 l}}_{\text{dissipation}}$$

We will think of barotropic pressure is $f(t)$ and barolinic is $f(t, z)$. We will write this gradient $\frac{\partial p}{\partial x}$ as P_x, the pressure gradient in the x-direction. We can now think about discretizing this equation,

$$\frac{\partial q^2}{\partial t} = \underbrace{\frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z})}_{n+1} + \underbrace{\nu_t \left(\frac{\partial u}{\partial z}\right)^2}_{n} + \underbrace{2\frac{g}{\rho_0}\kappa_p \frac{\partial p}{\partial z}}_{n} - \underbrace{2\frac{(q^2)^{3/2}}{B_1 l}}_{\text{mixed}}$$

Start to discretize...($Q = q^2$)

$$\frac{Q_i^{n+1} - Q_i^n}{\Delta t} = \frac{1}{\Delta z}\left(k_q \frac{Q_i^{n+1} - Q_{i-1}^{n+1}}{\Delta z}\right) + (\nu_t)_i^n \left(\left(\frac{\partial u}{\partial z}\right)_i^n\right)^2 + 2\frac{g}{\rho_0}(\kappa_p)_i^n \frac{\partial p}{\partial z} - 2\frac{(q^2)^{3/2}}{B_1 l}$$

We write the dissipation as a mixed discretization,

$$-2\left(\frac{\sqrt{Q}}{B_1 l}\right)_i^n \cdot Q_i^{n+1}$$

where it's partially linearized and partially explicit.

Phytoplankton Growth

We can write the population for phytoplankton C as

$$\frac{\partial C}{\partial t} + w_s \frac{\partial C}{\partial z} = \kappa \frac{\partial^2 C}{\partial z^2} + \Gamma(z) C \tag{9}$$

where $\Gamma(z)$ is the source + sink term (e.g., mortality + growth).

$$\Gamma(z) = p_i(I) - L_i$$

and $I(z,t)$ is the light intensity at depth z and time t. Thus, we expect Γ to vary as a function of depth and time.

We start to discretize the population for phytoplankton C as

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} + w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = \frac{\kappa}{\Delta z^2}\left(C_{i+1}^{n+1} - 2C_i^{n+1} + C_{i-1}^{n+1}\right) + \Gamma(z) C_i^{n+1} \tag{10}$$

$$-\frac{C_i^n}{\Delta t} = -\frac{C_i^{n+1}}{\Delta t} - w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} + \frac{\kappa}{\Delta z^2}\left(C_{i+1}^{n+1} - 2C_i^{n+1} + C_{i-1}^{n+1}\right) + \Gamma(z) C_i^{n+1}$$

Group by index,

$$-\frac{C_i^n}{\Delta t} = -\frac{1}{\Delta t} C_i^{n+1} - \frac{w_s}{\Delta z} C_i^{n+1} + \frac{w_s}{\Delta z} C_{i-1}^{n+1} + \frac{\kappa}{\Delta z^2} C_{i+1}^{n+1} - \frac{2\kappa}{\Delta z^2} C_i^{n+1} + \frac{\kappa}{\Delta z^2} C_{i-1}^{n+1} + \Gamma(z) C_i^{n+1}$$

$$-\frac{C_i^n}{\Delta t} = \left(-\frac{1}{\Delta t} - \frac{w_s}{\Delta z} - \frac{2\kappa}{\Delta z^2} + \Gamma(z)\right) C_i^{n+1} + \left(\frac{w_s}{\Delta z} + \frac{\kappa}{\Delta z^2}\right) C_{i-1}^{n+1} + \left(\frac{\kappa}{\Delta z^2}\right) C_{i+1}^{n+1} +$$

$$C_i^n = \left(\frac{\Delta t}{\Delta t} + \frac{w_s \Delta t}{\Delta z} + \frac{2\kappa \Delta t}{\Delta z^2} - \Delta t \Gamma(z)\right) C_i^{n+1} + \left(-\frac{w_s \Delta t}{\Delta z} - \frac{\kappa \Delta t}{\Delta z^2}\right) C_{i-1}^{n+1} + \left(-\frac{\kappa \Delta t}{\Delta z^2}\right) C_{i+1}^{n+1} +$$

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} = \frac{\kappa t}{\Delta z^2}\left(C_{i+1}^{n+1} - 2C_i^{n+1} + C_{i-1}^{n+1}\right)$$

And move all the terms for the next time step (n+1) to the RHS,

$$-C_i^n \Delta t = -\frac{C_i^{n+1}}{d} \frac{\kappa t}{\Delta z^2}\left(C_{i+1}^{n+1} - 2C_i^{n+1} + C_{i-1}^{n+1}\right)$$

The equation for the boundary condition is zero-flux,

$$w_s C + \kappa \frac{\partial C}{\partial z} = 0$$

$$w_s C_i^{n+1} + \kappa \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = 0$$

Monod equation for phytop

Huisman's equation:

$$\frac{\partial \omega_i}{\partial t} = p_i(I)\omega_i - L_i\omega_i + v_i \frac{\partial \omega_i}{\partial z} + D\frac{\partial^2 \omega_i}{\partial z^2} \tag{11}$$

They discretize the light dependence of phytoplankton growth using a Monod function

$$p_i(I) = p_{max,i} \frac{I}{H_i + I}$$

H_i is the half-saturation constant of light lmited growth and $p_{max,i}$ is the maximum specific production rate for species i.

Lambert-Beer's law applied to nonuniform population density distributions:

$$I(z,t) = I_{in} \exp\left(-\int_0^z \left(\sum_{j=1}^{n} k_j \omega_j(\sigma,t)\right) d\sigma - K_{bg} z\right) \tag{12}$$

where I_{in} is incident light intensity at the top of the water column, k_j is the specific light attenuation coefficient of the phytoplankton species j; K_{bg} is the background turbitiy (eg, not phytoplankton) and σ is an integration variable.

The boundary condition for phytoplankton at the top and bottom of the water column is zero-flux:

$$v_i \omega_i(z,t) + D\frac{\partial \omega_i}{\partial z} = 0 \mid_{z=0}$$

Problem 3

Write the equation of motion for a velocity fluctuation.

We start by recalling the equation of motion for momentum, after performing Reynold's decomposition and splitting components into their mean and fluctuating parts...

$$\partial_t(\bar{u_j} + u_j') + (\bar{u_i} + u_i')\partial_i(\bar{u_j} + u_j') = -\frac{1}{\rho}\partial_j(\bar{P} + P') + \nu\partial_k\partial_k(\bar{u_j} + u_j')$$

$$\partial_t \bar{u}_j + \partial_t u'_j + \bar{u}_i \partial_i \bar{u}_j + \bar{u}_i \partial_i u'_j + u'_i \partial_i \bar{u}_j + u'_i \partial_i u'_j = -\frac{1}{\rho}\partial_j \bar{P} - \frac{1}{\rho}\partial_j P' + \nu \partial_k \partial_k \bar{u}_j + \nu \partial_k \partial_k u'_j \quad (13)$$

Now we can subtract our RANS equation for momentum from this equation to isolate the turbulent components (u'). Recall our RANS equation for momentum reads

$$\partial_t \bar{u}_j + \bar{u}_i \partial_i \bar{u}_j = -\frac{1}{\rho}\partial_j \bar{P} + \nu \partial_k \partial_k \bar{u}_j - \partial_i \overline{u'_j u'_i}. \quad (14)$$

Subtracting Equation (2) from Equation (1) yields

$$\partial_t u'_j + \bar{u}_i \partial_i u'_j + u'_i \partial_i \bar{u}_j + u'_i \partial_i u'_j = -\frac{1}{\rho}\partial_j P' + \nu \partial_k \partial_k u'_j + \partial_i \overline{u'_j u'_i}. \quad (15)$$

$$\partial_t u'_j + \bar{u}_i \partial_i u'_j = -\frac{1}{\rho}\partial_j P' + \nu \partial_k \partial_k u'_j + \partial_i \overline{u'_j u'_i} - u'_i \partial_i \bar{u}_j - u'_i \partial_i u'_j. \quad (16)$$

The TKE Equation can be expressed as

$$\frac{\partial k}{\partial t} + \bar{U}_j \quad (17)$$

The density profile is calculated according to

$$\rho = \rho_0 (1 - \alpha (C - T_0)) \quad (18)$$

We start with the continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho U_i) = 0 \tag{19}$$

where flow is considered to be divergent-free, or solenoidal.

Meanwhile, the Reynolds-Averaged Navier Stokes reads:

$$\frac{\partial \overline{U}}{\partial t} + \overline{U_j}\frac{\partial \overline{U_k}}{\partial x_j} + \rho \epsilon_{jkl} f_k U_l = \frac{\partial}{\partial x_k}\overline{u_k u_j} - \frac{\partial \overline{P}}{\partial x_j} - g\hat{e_3} \tag{20}$$

The Boussinesq approximation is invoked to represent that density fluctuaions affect bouyancy but no other transport/state terms. These fluctuations are written $\beta\theta$, where β is the coefficient of thermal expansion.

Mellor and Herring (1973) uses the following equation:

$$\langle \frac{p}{\rho}(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i})\rangle = -\frac{q}{3l_1}(\overline{u_i u_j} - \frac{\delta_{ij}}{3}q^2) + C_1 q^2 (\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}) \tag{21}$$

We are establishing equations for the following: $\overline{u_k u_i u_j}$, $\overline{p u_j}$, $\overline{u_i u_j \theta}$, and $\overline{p\theta}$

Math Note The velocity gradient $\frac{\partial U_j}{\partial x_i}$ can be decomposed into symmetric and antisymmetric parths,

$$\frac{\partial U_j}{\partial x_i} = S_{ij} + \Omega_{ij}$$

where the symmetric rate-of-strain tensor is

$$S_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)$$

and the asymmetric part is,

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i}\right)$$

Parameterization

$$\langle u_k u_i u_j \rangle = \frac{3}{5}\ lqS_q \left(\frac{\partial \langle u_i u_j \rangle}{\partial x_k} + \frac{\partial \langle u_i u_k \rangle}{\partial x_j} + \frac{\partial \langle u_j u_k \rangle}{\partial x_i}\right) \tag{22}$$

In this equation S_q is a dimensionless number.

Sm :

$$\frac{A_1}{A_2}\frac{B_1(\gamma - C_1) - (B_1(\gamma_1 - C_1) + 6(A_1 + 3A_2))R_f}{B_1\gamma_1 - (B_1(\gamma_1 + \gamma_2) - 3A_1)R_f} \tag{23}$$

where

$$\gamma_1 = \frac{1}{3} - \frac{2A_1}{B_1}$$

and

$$\gamma_2 = \frac{B_2}{B_1} - \frac{6A_1}{B_1}$$

Sh:

$$S_h = \frac{A_2(1 - \frac{6A_1}{B_1})}{1 - (3A_2B_2G_h + 18A_1A_2G_h)} = \frac{A_2(1 - \frac{6A_1}{B_1})}{(1 - 3A_2G_h(B_2 + 6A_1))}$$

$$S_m = \frac{A_1(1 - 3C_1 - \frac{6A_1}{B_1}) + S_HG_h(18A_1 + 9A_1A_2)}{(1 - 9A_1A_2G_h)}$$

Richardson's number:

$$G_H = -\frac{L^2}{Q^2}\frac{g}{\rho_0}\frac{\partial}{\partial z}$$

Chosen parameters

$$C_1 = \gamma_1 - \frac{1}{3A_1}B_1^{1/3} \approx 0.08$$

$$B_2 = \frac{B_1^{1/3}}{P_{rt}}\frac{u_T^2\langle\theta^2\rangle}{H^2} \approx$$
$$B_1 = 16.6$$

writing equations based on code

$$bX = bX - \frac{aXcX}{bX}$$

$$dX = dX - \frac{aXdX}{bX}$$

$$u_* = U_{bed}\sqrt{C_d}$$

$$G_h = -\left(\frac{N_{BV} * L}{Q + Small}\right)^2$$

$$diss = 2\frac{dt * \sqrt{Q^2}}{B_1 L_p}$$

Derive equation for Reynold's Stress Transport

We multiply this equation by u'_k, expanding this into a 3x3 tensor equation.

$$u'_k \partial_t u'_j + u'_k \bar{u}_i \partial_i u'_j = -u'_k \frac{1}{\rho}\partial_j P' + u'_k \nu \partial_z \partial_z u'_j + u'_k \partial_i \overline{u'_j u'_i} - u'_k u'_i \partial_i \bar{u}_j - u'_k u'_i \partial_i u'_j. \tag{24}$$

Reynold's Average:

$$\langle u'_k \partial_t u'_j\rangle + \langle u'_k \bar{u}_i \partial_i u'_j\rangle = -\langle u'_k \frac{1}{\rho}\partial_j P'\rangle + \langle u'_k \nu \partial_z \partial_z u'_j\rangle + \langle u'_k \partial_i \overline{u'_j u'_i}\rangle - \langle u'_k u'_i \partial_i \bar{u}_j\rangle - \langle u'_k u'_i \partial_i u'_j\rangle. \tag{25}$$

The divergence of the averaged Reynold's Stress goes to zero, as it's an averaged quantity multiplied by a fluctuating value

$$\langle u'_k \partial_t u'_j\rangle + \langle u'_k \bar{u}_i \partial_i u'_j\rangle = -\langle u'_k \frac{1}{\rho}\partial_j P'\rangle + \langle u'_k \nu \partial_z \partial_z u'_j\rangle + \langle u'_k u'_i \partial_i \bar{u}_j\rangle - \langle u'_k u'_i \partial_i u'_j\rangle. \tag{26}$$

We now take a second equation, flip the indices

$$\langle u'_j \partial_t u'_k\rangle + \langle u'_j \bar{u}_i \partial_i u'_k\rangle = -\langle u'_j \frac{1}{\rho}\partial_k P'\rangle + \langle u'_j \nu \partial_z \partial_z u'_k\rangle + \langle u'_j u'_i \partial_i \bar{u}_k\rangle - \langle u'_j u'_i \partial_i u'_k\rangle \tag{27}$$

and add these together. To reduce the messiness, we will first look at the LHS,

$$\langle u'_j \partial_t u'_k \rangle + \langle u'_j \bar{u}_i \partial_i u'_k \rangle + \langle u'_k \partial_t u'_j \rangle + \langle u'_k \bar{u}_i \partial_i u'_j \rangle. \tag{28}$$

To simply, we apply the reverse product rule (ie, given we know)

$$\frac{\partial}{\partial t}\langle u'_k u'_j \rangle = \langle u'_k \frac{\partial}{\partial t} u'_j \rangle + \langle u'_j \frac{\partial}{\partial t} u'_k \rangle$$

and

$$\bar{u}_i \frac{\partial}{\partial x_i}\langle u'_k u'_j \rangle = \bar{u}_i \langle u'_k \frac{\partial}{\partial x_i} u'_j \rangle + \bar{u}_i \langle u'_j \frac{\partial}{\partial x_i} u'_k \rangle,$$

we can rewrite this LHS as

$$\partial_t \langle u'_k u'_j \rangle + \bar{u}_i \partial_i \langle u'_k u'_j \rangle.$$

Now onto the fun bit... the RHS. Currently we have

$$-\langle u'_k \frac{1}{\rho} \partial_j P' \rangle - \langle u'_j \frac{1}{\rho} \partial_k P' \rangle + \langle u'_k \nu \partial_z \partial_z u'_j \rangle + \langle u'_j \nu \partial_z \partial_z u'_k \rangle + \langle u'_k u'_i \partial_i \bar{u}_j \rangle + \langle u'_j u'_i \partial_i \bar{u}_k \rangle - \langle u'_k u'_i \partial_i u'_j \rangle - \langle u'_j u'_i \partial_i u'_k \rangle.$$

Some of this follows the same pattern (reverse product rule), allowing us to expand our fluctuating pressure term. Consider that

$$\frac{\partial}{\partial x_j}\langle u'_k P' \rangle = \langle u'_k \partial_j P' \rangle + \langle P' \partial_j u'_k \rangle.$$

We substitute

$$-\langle u'_k \frac{1}{\rho} \partial_j P' \rangle = \frac{1}{\rho}\left(\langle P' \partial_j u'_k \rangle - \partial_j \langle u'_k P' \rangle \right)$$

$$-\langle u'_j \frac{1}{\rho} \partial_k P' \rangle = \frac{1}{\rho}\left(\langle P' \partial_k u'_j \rangle - \partial_k \langle u'_j P' \rangle \right)$$

Next we identify the advection of mean flow by the Reynolds stress, i.e., the terms

$$\langle u'_k u'_i \partial_i \bar{u}_j \rangle + \langle u'_j u'_i \partial_i \bar{u}_k \rangle.$$

We won't rearrange these terms for now, since they're already in a form involving an averaged Reynold's stress. The viscous dissipation, however, looks like an interesting challenge,

$$\nu \langle u'_k \partial_z \partial_z u'_j \rangle + \nu \langle u'_j \partial_z \partial_z u'_k \rangle.$$

We would like to write this as the diffusion of the Reynolds stress. We expect such diffusion to look like

$$\nu \partial_z \partial_z \langle u'_k u'_j \rangle,$$

which would expand (ignoring the coefficient) as

$$\partial_z \left(\partial_z \langle u'_k u'_j \rangle \right) = \partial_z (\langle u'_k \partial_z u'_j \rangle + \langle u'_j \partial_z u'_k \rangle)$$

$$= \langle u'_k \partial_z \partial_z u'_j \rangle + \langle \partial_z u'_j \partial_z u'_k \rangle + \langle \partial_z u'_j \partial_z u'_k \rangle + \langle u'_j \partial_z \partial_z u'_k \rangle.$$

We relate this to the term in the equation by substituting

$$\nu \langle u'_k \partial_z \partial_z u'_j \rangle + \nu \langle u'_j \partial_z \partial_z u'_k \rangle = \nu \partial_z \partial_z \langle u'_k u'_j \rangle - 2\nu \langle \partial_z u'_j \partial_z u'_k \rangle.$$

The last terms we want to deal with are the advection of turbulent fluctuations by the Reynolds stress itself,

$$-\langle u'_k u'_i \partial_i u'_j \rangle - \langle u'_j u'_i \partial_i u'_k \rangle.$$

We recognize this pattern, and can apply reverse-product-rule. The last step is due to impressibility (allows us to put u_i inside the derivative).

$$-\langle u'_i u'_k \partial_i u'_j \rangle - \langle u'_i u'_j \partial_i u'_k \rangle = -\langle u'_i \partial_i u'_k u'_j \rangle = -\langle \partial_i u'_k u'_j u'_i \rangle$$

Now we can put together all the steps. For notation in this last step, I am using capital U to denote an averaged velocity (instead of an overbar).

$$\partial_t \langle u'_k u'_j \rangle + U_i \partial_i \langle u'_k u'_j \rangle = \tag{29}$$

`% triple correlation`

$$-\partial_i \langle u'_k u'_j u'_i \rangle + \langle u'_k u'_i \rangle \partial_i U_j + \langle u'_j u'_i \rangle \partial_i U_k$$

`% viscous stress`

$$+\nu \partial_z \partial_z \langle u'_k u'_j \rangle - 2\nu \langle \partial_z u'_j \partial_z u'_k \rangle + \frac{1}{\rho} \left(P' \langle \partial_j u'_k + \partial_k u'_j \rangle - \partial_j \langle u'_k P' \rangle - \partial_k \langle u'_j P' \rangle \right)$$

We can define all these terms and define them from a physics perspective. Starting left to right,

1. Storage / time variation of the Reynold's stress
2. Advection of the Reynold's stress by the mean flow
3. Triple Correlation: transport of Reynold's stress by turbulent motions.
4. Advection of flow by Reynold's stress. This is also seen as a production term, as it captures the conversion of turbulence from the mean flow through shear.
5. (shear production)
6. Diffusion of mometum from viscosity / molecular interactions.
7. This is a very exciting term! It is the viscous dissipation term and it *directly destroys* Reynold's stress through conversion to heat.
8. These next two terms are very interesting. Franko spent a long time talking to me about this one before he graduated, and said if I understood this term, I would "understand some grand part of turbulence." Whether or not that is true has yet to be seen. However, this is the pressure-strain correlation. It resembles the rate-of-strain tensor, and similar to S_{ij}, this tensor is traceless. This term cannot create or destroy turbulence, however, it often performs what is called "pressure scrambling." One of these terms will anisotropize the turbulence ("fast term") and the other isotropizes the flow ("slow term").
9. These terms are transport by a pressure-fluctuation correlation term.